

Multimodal AI Framework for Personalized and Culturally-Aware Skin Disease Diagnosis, Monitoring, and Treatment Support

1. Introduction

1.1 Purpose

This document presents a comprehensive Software Requirement Specification (SRS) for the development of a Multimodal AI Framework for Skin Disease Diagnosis, Monitoring, and Treatment Support. The purpose of this system is to design an intelligent healthcare support solution that enhances the accuracy and reliability of skin disease diagnosis by integrating multiple data sources, including dermatological images, patient-reported symptoms in text and audio formats, and clinical information.

The system aims to address key limitations in traditional and existing AI-based dermatology systems, such as dependency on image-only analysis, lack of contextual understanding, and bias across different skin tones. In addition, the system will provide personalized treatment recommendations and disease monitoring capabilities, making it a comprehensive decision-support tool for both patients and healthcare professionals.

1.2 Scope

The proposed system is a web-based application designed to assist users in diagnosing skin diseases and receiving treatment guidance. It allows patients to upload skin images, describe symptoms through text or voice, and receive accurate predictions along with recommendations.

The system is intended to:

- Improve diagnostic accuracy using multimodal data
- Provide early detection and monitoring of skin diseases
- Reduce dependency on dermatologists in underserved regions
- Ensure fairness and inclusivity across different populations
- Offer personalized treatment support based on patient-specific factors

The system is not intended to replace medical professionals but to act as a decision-support system that assists in clinical evaluation.

1.3 Definitions and Abbreviations

- AI – Artificial Intelligence
 - CNN – Convolutional Neural Network
 - NLP – Natural Language Processing
 - ASR – Automatic Speech Recognition
 - LLM – Large Language Model
 - SRS – Software Requirement Specification
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2. Overall Description

2.1 System Overview

The proposed system is a multimodal AI-based platform that integrates image analysis, text processing, and audio processing to provide a more accurate and context-aware diagnosis of skin diseases. Unlike traditional systems that rely solely on visual data, this system incorporates patient symptoms and clinical information to improve prediction reliability.

2.2 System Functions

The system performs several core functions:

- Accept and process dermatological images
 - Capture and analyze patient symptoms in text and audio formats
 - Extract meaningful features from multiple data sources
 - Combine multimodal inputs using fusion techniques
 - Classify skin diseases with high accuracy
 - Provide Top-3 predictions with confidence scores
 - Monitor disease progression over time
 - Generate personalized treatment recommendations
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2.3 User Classes

The system is designed for multiple types of users:

- Patients: Individuals who use the system to upload images, provide symptoms, and receive diagnosis and recommendations.
 - Dermatologists: Medical professionals who can review predictions and use the system as a support tool.
 - Administrators: Responsible for system maintenance, updates, and data management.
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2.4 Operating Environment

The system operates as a web-based platform and can be accessed through:

- Desktop computers
- Mobile devices
- Web browsers

It requires a stable internet connection and supports integration with backend AI models and databases.

3. System Features

3.1 Image Processing Module

This module handles the preprocessing and analysis of skin images. It performs operations such as resizing, normalization, and augmentation to improve data quality. Deep learning models such as CNNs are used to extract features and classify diseases.

3.2 Symptom Analysis Module

This module processes patient-reported symptoms in both text and audio formats. NLP techniques analyze textual input, while ASR converts speech into text. This helps the system understand contextual information beyond visual features.

3.3 Multimodal Fusion Module

The multimodal fusion module integrates outputs from image and symptom analysis models. Fusion techniques such as weighted averaging and attention mechanisms are used to generate a more accurate and reliable prediction.

3.4 Classification Module

This module classifies skin diseases into predefined categories using trained machine learning models. It provides both Top-1 and Top-3 predictions along with confidence scores.

3.5 Disease Monitoring Module

This module tracks disease progression over time by analyzing sequential images and data. It helps determine whether the condition is improving, worsening, or stable.

3.6 Treatment Recommendation Module

This module generates personalized treatment suggestions based on diagnosis and patient-specific factors. Knowledge graphs and LLMs are used to provide safe, explainable, and context-aware recommendations.

4. Functional Requirements

- Image Processing – The system must accept dermatology images and perform preprocessing such as resizing, normalization, and augmentation to prepare inputs for AI models.
- Audio Input Handling – The system must allow users to record and submit audio descriptions of symptoms in multiple languages.
- Speech-to-Text Conversion – The system must convert audio inputs into text using ASR technologies.
- Text Processing – The system must analyze symptom descriptions using NLP models.
- Feature Extraction – The system must extract relevant features from image, text, and audio inputs.
- Multimodal Fusion – The system must integrate multiple data types to generate accurate predictions.
- Disease Classification – The system must classify skin diseases into predefined categories.
- Top-3 Predictions – The system must provide multiple possible diagnoses for better reliability.
- Confidence Scores – The system must display prediction confidence levels.
- Disease Monitoring – The system must track disease progression over time.
- Treatment Recommendation – The system must generate personalized treatment suggestions.

- User Interface – The system must provide an easy-to-use interface for interaction.
 - Data Storage – The system must securely store user data and results.
 - Feedback Generation – The system must provide clear and understandable outputs.
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5. Non-Functional Requirements

- Accuracy – The system must achieve high accuracy in predictions using multimodal data.
 - Performance – The system must provide fast response times for all operations.
 - Scalability – The system must handle increasing users and data efficiently.
 - Security & Privacy – The system must ensure secure storage and protection of user data.
 - Reliability – The system must function consistently with minimal downtime.
 - Usability – The system must be user-friendly and easy to navigate.
 - Robustness – The system must handle low-quality or incomplete inputs.
 - Fairness – The system must perform equally across different skin tones and populations.
 - Maintainability – The system must be easy to update and maintain.
 - Compatibility – The system must work across multiple devices and platforms.
 - Explainability – The system must provide understandable explanations for predictions.
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6. External Interface Requirements

6.1 User Interface

The system provides a simple and interactive interface where users can upload images, input symptoms, and view results.

6.2 Hardware Interface

- Camera for capturing images
- Microphone for recording audio

6.3 Software Interface

- AI models (CNN, NLP, ASR)
- Database systems

- REST APIs for integration
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7. System Architecture

The system follows a modular architecture consisting of:

- Frontend (User Interface)
- Backend (API and server)
- AI Processing Modules
- Database

Each module communicates through APIs to ensure efficient data flow and system performance.

8. Future Enhancements

- Mobile application development
 - Real-time monitoring features
 - Integration with hospital systems
 - Cloud deployment
 - Expansion of datasets
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9. Conclusion

The proposed system provides a comprehensive and intelligent solution for skin disease diagnosis by integrating multiple data sources and advanced AI techniques. It improves accuracy, fairness, and usability, making it a valuable tool for modern healthcare systems.